An Explainable Multi-Mode Board Game Recommendation System Using Content Similarity, Sentiment Analysis, and Popularity Signals

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***Abstract*—BoardGameGeek (BGG) lists more than 21,000 board games, and players who want something new have to search a large catalog with little guidance. Collaborative filtering needs rating histories that new players and new games do not have, so we built a recommender that works from game content instead and explains each suggestion. It offers three modes: a title-based mode that starts from games the user already likes, a trait-based mode that starts from categories and mechanics, and a combined mode that uses both. The scores combine TF-IDF and sentence-embedding similarity, BGG attribute overlap, review sentiment from a DistilBERT classifier, and popularity, and every recommendation lists the attributes it shares with the user's input.**

We evaluated the system on 300 BGG users with Recall@K and NDCG@K under a holdout protocol, together with a popularity baseline, a content-only baseline, and a sentiment ablation. The held-out games in this protocol are mostly the most-rated titles, and the popularity baseline won by a wide margin (Recall@10 of 0.230). The three proposed modes reached Recall@10 between 0.020 and 0.037, with the combined mode best. On held-out games outside the 100 most-rated titles, the popularity baseline retrieved none and the proposed modes retrieved a small share (Recall@10 of 0.015–0.022). Removing sentiment made no measurable difference.

On this data, offline accuracy depends mostly on popularity. The value of the framework lies in explainable, similarity-based discovery beyond the best-known games, and showing that value will take a user study and an evaluation that leans less on popular titles.

***Index Terms*—Recommender Systems, Board Game Recommendation, Explainable Recommendation, Content-Based Filtering, Sentiment Analysis, Hybrid Recommendation, BoardGameGeek**

I. Introduction

BoardGameGeek (BGG), an online board game database, lists tens of thousands of titles. A player who enjoyed one game and wants another like it has to search that catalog by hand or rely on lists of popular titles. A recommender can help, but it has to cope with a catalog where most games have relatively few ratings and many players have rated only a handful.

Collaborative filtering recommends items from patterns in user ratings and works well when those patterns are dense. It struggles with the cold-start problem: a new game, or a user with a short rating history, gives the method little to work with. Sparse ratings also make its predictions less reliable for the many games in the long tail of the catalog.

Content-based methods avoid the cold-start problem by comparing the items themselves, here through their descriptions, mechanics, categories, and themes. They miss what players say about a game, however, and popularity-based lists favor well-known titles over the niche games a player might enjoy.

We combine these signals in one system and keep it explainable. The system scores games by content similarity, attribute overlap, review sentiment, and popularity, and it tells the user which attributes connect each recommendation to their input. It offers title-based, trait-based, and combined modes, and we measure how much each signal contributes.

This paper makes four contributions:

- A recommendation framework that combines content similarity, attribute overlap, review sentiment, and popularity.
- An explanation for every recommendation, built from the BGG attributes that link it to the user's input.
- Three query modes: title-based, trait-based, and combined.
- An evaluation with Recall@K and NDCG@K that includes popularity and content-only baselines, a sentiment ablation, and a long-tail analysis.

Section II reviews related work. Section III describes the data, the features, and the scoring. Section IV presents the evaluation and its results, Section V discusses them, and Section VI concludes.

II. Related Work

Collaborative filtering is one of the most established recommendation approaches. It finds users or items with similar rating patterns and recommends accordingly. Su and Khoshgoftaar [1] survey collaborative filtering techniques and describe their main weaknesses, rating sparsity and cold start among them. Herlocker et al. [2] discuss how to evaluate collaborative filtering systems properly.

Content-based systems recommend items with attributes similar to those a user liked, so they do not depend on other users' histories. Pazzani and Billsus [3] describe how such systems model item similarity from text and metadata.

Board games suit this approach because each game carries a description, mechanics, categories, themes, and a complexity rating. Metadata cannot capture everything players care about, though, such as how a game feels to play.

Hybrid systems combine several sources of evidence to offset the weaknesses of each. Burke [4] presents hybrid recommenders as a way to reduce the weaknesses of individual approaches, and a more recent survey [5] shows that hybrid designs remain common across domains because they can mix collaborative, content-based, and contextual signals. Our system follows this line by combining content similarity, sentiment, and popularity.

Sentiment-aware recommenders draw on the opinions in user reviews. Wang et al. [6] combine review sentiment with contextual signals, and Da'u and Salim [7] report that sentiment-aware models improve recommendation performance by using opinion information from review text. Board game reviews often comment on enjoyment, difficulty, and replay value, which ratings alone do not convey.

Explainable recommendation has become an active research area. Zhang and Chen [8] argue that explanations improve user trust and support decisions, and Chatti et al. [9] study how visualization can explain recommendations. A board game player benefits from knowing why a game was suggested, for example because it shares mechanics or themes with a favorite.

Ranking metrics such as Recall@K and NDCG@K measure whether relevant items appear near the top of a list. Radlinski and Craswell [10] compare the sensitivity of information retrieval metrics, and Sakai and Kando [11] examine metrics under incomplete relevance judgments. Both support ranking-based evaluation when the order of results matters.

Prior work leaves a gap for board games. Collaborative filtering struggles with sparse data, pure content-based systems ignore what players say, and popularity lists favor mainstream titles. Few systems combine flexible query modes with explanations an ordinary user can read. We address that gap with a system that integrates content similarity, sentiment, and popularity and supports title-based, trait-based, and combined queries.

III. Methodology

This section covers the data, the features, the scoring in each mode, and the evaluation protocol.

A. Dataset Description

We used public BoardGameGeek data: game metadata, user ratings, and the review comments that accompany some ratings. The game catalog contains the 21,631 games in the metadata file that have a unique, valid identifier. For each game we kept its description, categories, mechanics, families, publishers, complexity rating (BGG average weight), and rating statistics.

We restricted the user ratings to games in the catalog, which left 18,744,924 ratings (approximately 18.7 million) from 410,471 users.

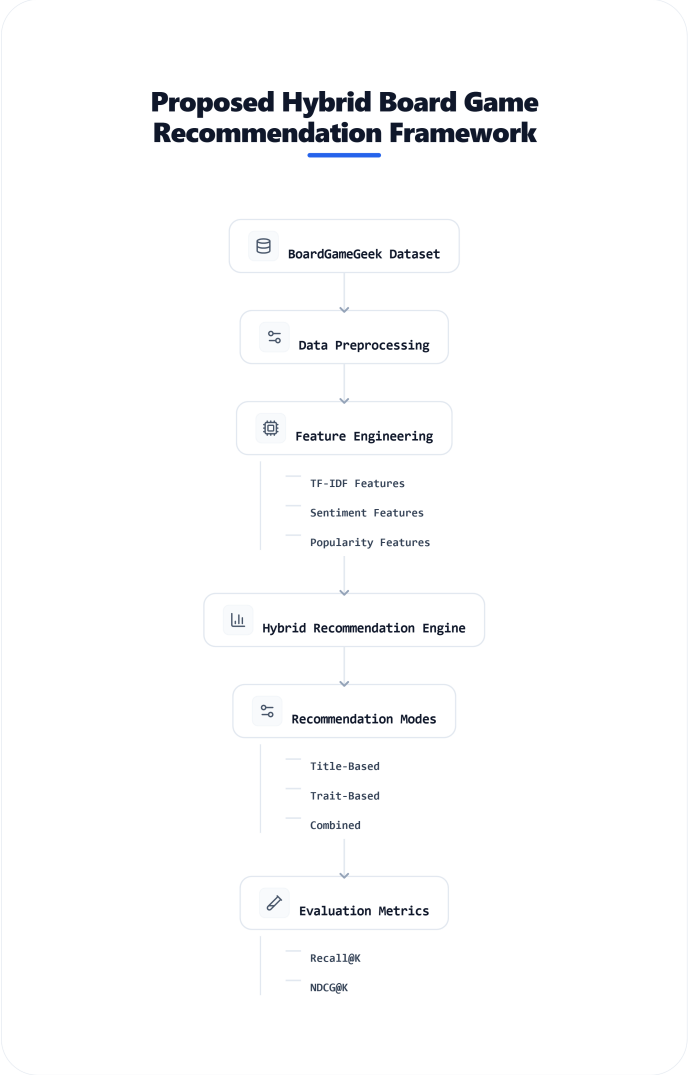


Fig. 1. Overview of the recommendation framework

For evaluation, we excluded users with fewer than six rated games and split each remaining user's ratings into seed games and held-out targets, as Section III-G describes.

The review comments feed the sentiment signal. The descriptions and attributes feed the similarity signals.

B. Feature Engineering

We derived three kinds of signal from the metadata and reviews: content similarity, sentiment, and popularity.

1) Content Representation: We computed content similarity from three representations of each game:

- **Description TF-IDF**: unigram and bigram TF-IDF vectors (20,000 features) of the BGG game description, which capture keyword-level similarity.
- **Attribute vectors**: L2-normalized binary indicators of the game's BGG mechanics, themes, and subcategories, which capture similarity in how a game plays.

- **Semantic embeddings**: the game name and description encoded with the all-MiniLM-L6-v2 sentence-transformer (first 256 tokens), which capture semantic relationships that keyword overlap misses.

Within each representation we computed cosine similarity and stored each game's 50 nearest neighbors in a sparse matrix.

2) Trait Extraction: The trait-based score uses four structured BGG attributes: categories, mechanics, families, and publishers. For each attribute, a candidate game's overlap score is the fraction of the user's requested values that the game has. A small hand-written expansion table maps broad keywords to related BGG terms, so users can describe what they want in everyday words. For example, "strategy" expands to categories such as Economic and Territory Building and mechanics such as Hand Management and Tile Placement.

Trait matching is a soft overlap, not a strict filter: a game that has more of the requested traits receives a higher score. Complexity (BGG average weight) does not enter the trait score. The user can instead apply it as an optional filter after ranking, with three bands (low ≤ 2.0, medium 2.0–3.0, high > 3.0).

3) Sentiment Analysis: We scored review sentiment with the DistilBERT SST-2 classifier from the HuggingFace transformers pipeline.

The review corpus holds about 26 million ratings, too many to classify on a CPU. We therefore sampled up to ten non-empty comments per game uniformly at random, which gave 209,162 comments across 21,581 catalog games, and truncated each comment to 128 tokens. The classifier returns a label and its confidence in that label, so we converted each prediction to the probability that the review is positive (p for a POSITIVE label, $1 - p$ for a NEGATIVE one) and averaged these per game. Games with fewer than three scored comments received no score and fall back to the catalog mean, which leaves 21,510 games with their own sentiment value. We min-max normalized the per-game values before adding them to the hybrid score.

4) Popularity Signal: Popularity comes from BGG's rating statistics. In the title-based score, popularity P is the game's number of user ratings, min-max normalized over the catalog. The trait-based and combined modes also use the Bayesian average rating (R) and the log-scaled rating count (V), both min-max normalized over the candidate set, as small tie-breaking terms.

We normalized the popularity features before combining them so that they share the $[0, 1]$ scale of the similarity signals.

C. Score Normalization

The signals start on different scales, so we normalized each one before combining them.

For a query with one or more seed games, a candidate's similarity in each matrix is the mean of its stored similarities to the seeds that list it among their 50 neighbors, and zero if none do. The keyword signal T is the average of the description and attribute similarities, and the semantic signal E is the embedding similarity. We min-max normalized both over the whole catalog. Trait overlap scores already lie in $[0, 1]$ because they are proportions of matched attributes.

We also min-max normalized the rating counts and the per-game sentiment scores before they enter the formulas below.

D. Hybrid Recommendation Formula

Each mode combines the normalized signals with its own fixed weights. The title-based score for a candidate game is

$$S_{title} = 0.35\,T + 0.35\,E + 0.10\,S + 0.20\,P \tag{1}$$

where T is the keyword (description and attribute) similarity, E the semantic embedding similarity, S the normalized sentiment score, and P the normalized rating count.

The trait overlap score combines the four attribute overlaps (categories O_{cat}, mechanics O_{mech}, families O_{fam}, publishers O_{pub}):

$$S_{trait} = 0.40\,O_{cat} + 0.35\,O_{mech} + 0.15\,O_{fam} + 0.10\,O_{pub} \tag{2}$$

and the trait-based mode ranks candidates by $0.70\,S_{trait} + 0.15\,R + 0.10\,V + 0.05\,S$. The combined mode merges both candidate sets and ranks by

$$S_{comb} = 0.65\,S_{title} + 0.35\,S_{trait} + 0.08\,R + 0.04\,V + 0.03\,S \tag{3}$$

where a candidate missing from one side contributes zero for that side.

We set the weights by hand and did not learn or tune them on the evaluation data. Similarity terms carry most of the weight because the system's purpose is to find games like the ones a user names; popularity and sentiment play a supporting part.

E. Recommendation Modes

The three modes cover different starting points.

1) Title-Based Recommendation: The user names one or more games they like, and the system ranks the catalog by S_{title}. Content similarity drives this mode, with sentiment and popularity in a supporting role.

2) Trait-Based Recommendation: The user names categories, mechanics, families, or publishers, optionally with a complexity band, and the system ranks games by how many of those traits they have. This mode serves users who have no particular game in mind.

3) Combined Recommendation: The user gives both seed games and traits, and the system ranks candidates by S_{comb}. Games that resemble the seeds and also match the requested traits rank highest.

F. Post-Ranking Rules

Two rules run after ranking and therefore change no score. The optional difficulty filter keeps only games in the selected complexity band. The same-series cap keeps at most two results that share a BGG series tag (family tags beginning with "Game:" or "Series:") with a seed game. Without the cap, pure similarity fills the list with a seed's own editions and expansions, for example six Ticket to Ride variants for a Ticket to Ride seed, and leaves little room for new discoveries.

The explanation names the shared series, so the user can see why the rule applied.

G. Evaluation Strategy

We evaluated the framework with a holdout protocol built from the BGG rating data (18.7 million ratings on catalog games).

Of the 259,737 users with at least six rated games, we evaluated the first 300 in the order produced by grouping the ratings by username. For each user, the first three rated games in the order they appear in the ratings file served as seed games, and the next three were held out as relevance targets. The ratings file lists games with the most ratings first, so both the seeds and the held-out games lean toward popular titles. This matters when comparing against a popularity baseline.

We measured Recall@K and NDCG@K at several cutoffs. A recommendation counts as relevant if it matches one of the user's held-out games, whatever rating the user gave it. For the trait-based and combined modes, we derived the trait query from the seeds: their two most frequent categories and two most frequent mechanics. All modes ran through the same engine code that serves the deployed application.

Recall@K measures the share of relevant games that appear in the top K, and NDCG@K rewards placing them near the top of the list.

H. System Implementation

We implemented the system in Python. Pandas and NumPy handle the data, scikit-learn computes the TF-IDF vectors and similarities, and a Streamlit application lets users run queries and read the explanations.

The code keeps the scoring engine separate from the interface, so the same engine serves both the application and the evaluation notebook.

IV. Evaluation

This section shows example explanations, defines the metrics, and reports the results.

A. Explainability Example

The system attaches an explanation to every recommendation. The explanation lists the BGG attributes behind the match: the categories, mechanics, families, and publishers that the recommended game shares with the seed games (title-based mode) or with the requested traits (trait-based and combined modes). The interface also shows the component scores (S_{title}, S_{trait}, and the final score), the Bayesian average rating, and the rating count, so users can see how much of a ranking comes from similarity and how much from the supporting signals. Because title lookup falls back to substring matching, the interface also reports which catalog game each typed title matched.

Table I reproduces, verbatim, the explanations the system produced for three example queries.

TABLE I
VERBATIM EXPLANATIONS PRODUCED BY THE SYSTEM (PUBLISHER LISTS ABBREVIATED)

Query → Game	System explanation
Splendor + Azul → Sultaniya	Categories: Puzzle | Mechanics: Pattern Building, Set Collection, Tile Placement
Splendor + Azul → Azul: Summer Pavilion	Categories: Abstract Strategy, Puzzle | Mechanics: Card Drafting, End Game Bonuses, Pattern Building, Set Collection, Tile Placement, Turn Order: Claim Action | Families: Game: Azul | Publishers: Broadway Toys LTD, ...
Ticket to Ride → Railroad Ink: Deep Blue Edition	Categories: Trains | Mechanics: Network and Route Building | Publishers: Edge Entertainment, Galápagos Jogos
Horror + Mystery (traits) → Mysterium	Categories: Deduction, Horror | Mechanics: Cooperative Game, Deduction

B. Evaluation Metrics

We use two standard ranking metrics from recommender systems research: Recall@K and Normalized Discounted Cumulative Gain (NDCG@K).

Recall@K is the share of a user's relevant items that appear among the top-K recommendations:

$$Recall@K = \frac{\text{Number of Relevant Recommended Items}}{\text{Total Number of Relevant Items}} \tag{4}$$

NDCG@K also accounts for position, giving more credit to relevant items near the top of the list. The Discounted Cumulative Gain (DCG) is

$$DCG@K = \sum_{i=1}^{K} \frac{rel_i}{\log_2(i+1)} \tag{5}$$

and its normalized form is

$$NDCG@K = \frac{DCG@K}{IDCG@K} \tag{6}$$

where $IDCG@K$ is the DCG of an ideal ranking that places every held-out item first.

We chose these metrics because users of a discovery tool tend to look only at the first few results.

C. Experimental Setup

We evaluated six models on the same 300 users:

- Title-based recommendation,
- Trait-based recommendation,
- Combined hybrid recommendation,
- a popularity baseline,
- a content-only baseline, and
- a title-based variant without sentiment (ablation).

The three proposed modes used the scoring described in Section III, with no difficulty filter. Each result comes from a single run, without repeated sampling or significance testing.

1) Popularity-Based Baseline: The popularity baseline recommends the games with the most user ratings, excluding the user's seed games, regardless of what the seeds are. Many platforms use this simple strategy.

2) Content-Based Baseline: The content-based baseline is the title-based scorer with the sentiment and popularity weights set to zero ($0.5\,T + 0.5\,E$). It shows what content similarity achieves on its own, and therefore what the other signals add.

3) Sentiment Ablation: To isolate the sentiment signal, we also ran the title-based scorer with only the sentiment weight set to zero ($0.35\,T + 0.35\,E + 0.20\,P$).

D. Experimental Results

Table II reports the main protocol, and Fig. 2 plots Recall@10 and NDCG@10.

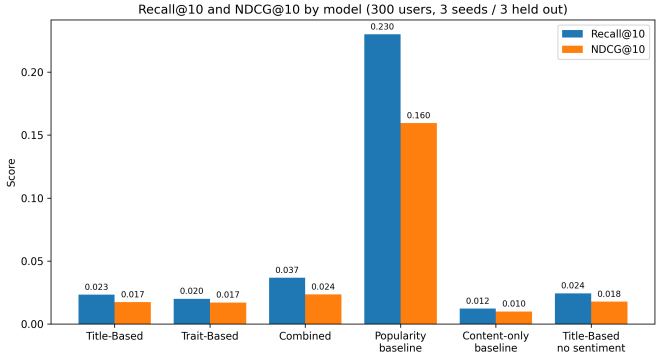


Fig. 2. Recall@10 and NDCG@10 of the proposed modes, the baselines, and the sentiment ablation (300 users)

TABLE II
RECOMMENDATION PERFORMANCE (300 USERS, 3 SEEDS / 3 HELD OUT)

Model	Recall@5	NDCG@5	Recall@10	NDCG@10	Recall@20	NDCG@20
Title-Based	0.0133	0.0128	0.0233	0.0174	0.0411	0.0236
Trait-Based	0.0133	0.0140	0.0200	0.0171	0.0444	0.0256
Combined	0.0222	0.0168	0.0367	0.0235	0.0556	0.0302
Popularity baseline	0.1400	0.1190	0.2300	0.1595	0.3622	0.2068
Content-only baseline	0.0111	0.0095	0.0122	0.0100	0.0167	0.0116
Title-Based, no sentiment	0.0133	0.0128	0.0244	0.0178	0.0367	0.0222

The popularity baseline outperformed all three proposed modes by a wide margin, with a Recall@10 of 0.230 against 0.037 for the best proposed mode (combined). The protocol explains much of this gap. Of all held-out games, 63.0% are among the 100 most-rated titles in the catalog, so a list of the most-rated games retrieves many of them by construction. Among the proposed modes, the combined mode scored highest on every metric, and the title-based and trait-based modes scored about the same.

Table III scores the same ranked lists against only the held-out games outside the 100 most-rated titles, for the 155 users who have at least one such game. We fixed the cut-off before looking at the results. The popularity baseline retrieves none of these games, since its lists stay among the most popular titles, while the proposed modes retrieve some less popular games that users went on to rate. The absolute values are low, so the table shows a difference in behavior rather than strong long-tail accuracy.

TABLE III
LONG-TAIL ANALYSIS: HELD-OUT GAMES OUTSIDE THE 100 MOST-RATED (155 USERS)

Model	Recall@5	NDCG@5	Recall@10	NDCG@10	Recall@20	NDCG@20
Title-Based	0.0129	0.0082	0.0151	0.0092	0.0258	0.0124
Trait-Based	0.0054	0.0051	0.0183	0.0093	0.0344	0.0141
Combined	0.0194	0.0101	0.0215	0.0110	0.0398	0.0166
Popularity baseline	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Content-only baseline	0.0129	0.0090	0.0151	0.0100	0.0215	0.0122
Title-Based, no sentiment	0.0129	0.0082	0.0151	0.0091	0.0280	0.0129

The two comparison models isolate the supporting signals. Removing only the sentiment term changed the title-based Recall@10 from 0.0233 to 0.0244, a difference of a single hit across the 900 held-out games, so sentiment has no measurable effect in this evaluation. Removing both sentiment and popularity (the content-only baseline) lowered Recall@10 to 0.0122. This fits the popularity bias of the protocol: the popularity term, more than content similarity, accounts for the title-based mode's offline accuracy.

V. DISCUSSION

On this dataset and protocol, popularity predicts which games a user rates next far better than content similarity does. Offline recommender evaluations often show this popularity bias. BGG users rate many of the same widely played titles, and a protocol that holds out a user's next ratings in a popularity-ordered file inherits that bias. An earlier version of this system appeared to reach a title-based Recall@10 of 0.20. That figure was an artifact: we had computed the semantic-embedding and sentiment signals for only 20 very popular games, so the title-based mode recommended those 20 titles to almost everyone. Once we computed both signals for the full catalog, the title-based mode recommended similar games and its accuracy fell to the level reported here.

We designed the proposed modes for discovery, not for predicting the next popular game a user rates. The long-tail analysis supports this distinction, since only the similarity-based modes reach games outside the 100 most popular, but it does not show that they reach them accurately. Combining seed similarity with explicit traits gave the best results among the proposed modes in the main analysis, which suggests that a user's stated preferences add information that seed similarity alone lacks.

The explanations are the system's main strength. Each recommendation comes with the categories, mechanics, families, and publishers it shares with the user's input (Table I), along with the component scores and the catalog game each typed title matched. Their main weakness is the publisher list: popular games carry many publishers of localized editions, and listing them all adds noise to an otherwise informative explanation.

The study has several limitations. First, the evaluation uses one protocol with a popularity-ordered split, treats every rated

game as relevant regardless of its rating, covers only 300 users, and reports single runs without significance testing. Second, sentiment comes from at most ten sampled comments per game, so each game's value is noisy. Third, we set the weights by hand and did not tune them. Fourth, the semantic embeddings include the game name, which gives games with similar names a small similarity boost. Finally, the system does not use collaborative filtering or model individual users.

Future work should evaluate the system with a randomized split and a rating threshold for relevance, and report confidence intervals. A user study that measures discovery and the usefulness of the explanations directly would test the system's main claim. Learning the weights on a validation split, re-ranking for diversity beyond the series cap, and adding a collaborative component to the content signals are further extensions.

VI. Conclusion

We presented an explainable board game recommender with three query modes that combines content similarity, attribute matching, review sentiment, and popularity, and explains each recommendation through the attributes behind it. We evaluated it through the same code that serves the deployed application. On a holdout protocol dominated by popular games, a popularity baseline outperformed all three modes by a wide margin (Recall@10 of 0.230 against 0.037 for the best mode), while the modes retrieved a small share of less popular held-out games that the baseline never recommends.

The results do not support a claim of superior retrieval accuracy. They support a narrower one: a lightweight, interpretable, content-based framework can surface relevant games beyond the most popular titles and say why it chose them. Whether that helps players discover games remains to be shown with user studies and with evaluation that depends less on predicting the next rating.

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